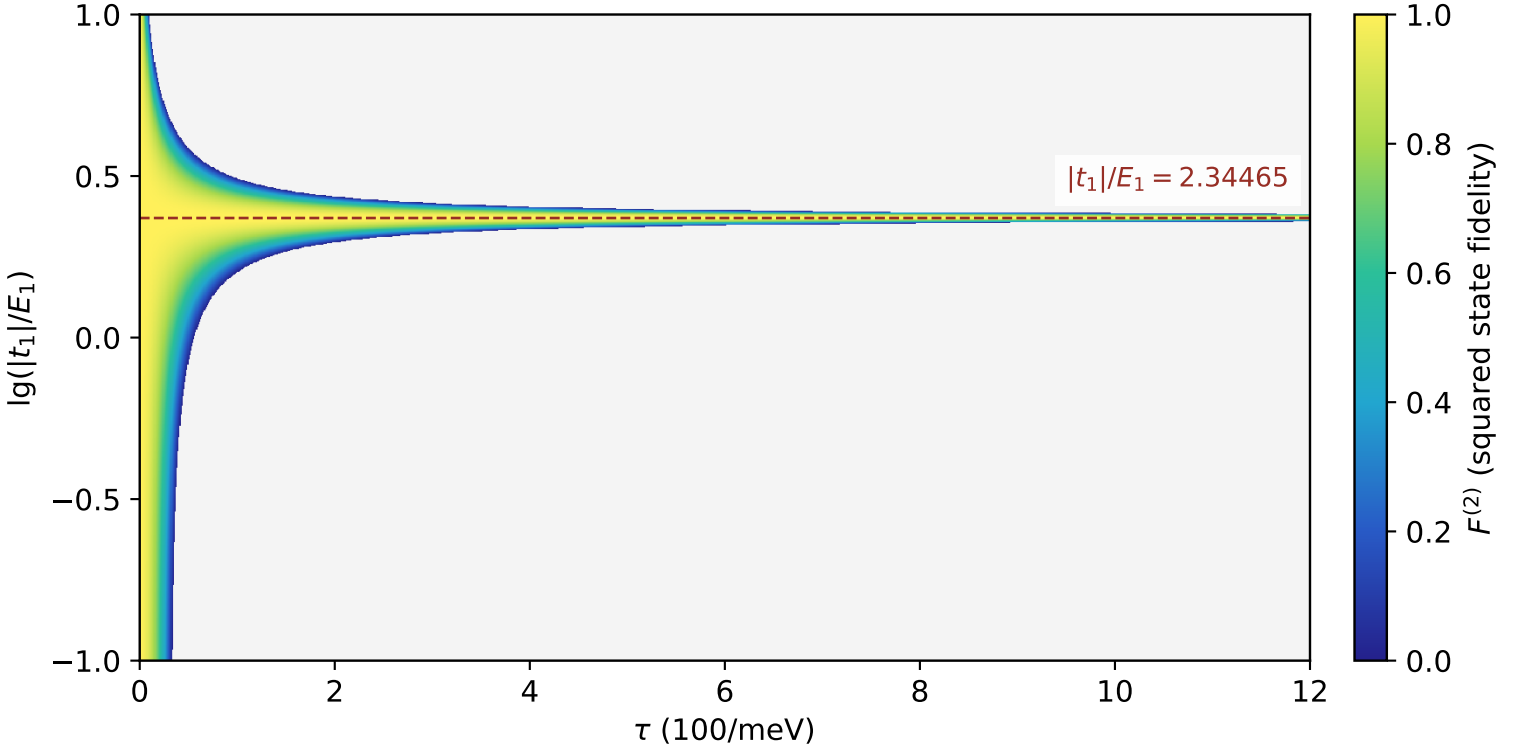
Derived SO(5) fidelity formula on Fig. 1(d) axes

$$F^{(2)} = 1 - 2(\tau/\hbar)^2[2.180340599\,E_1 - 0.929923300\,|t_1|]^2$$



$E_1 = 0.01$ meV; equal ideal control amplitudes; $\hbar = 1$; two three-step exchanges.

Blank region: $F^{(2)} < 0$ (unphysical extrapolation). Dashed line: quadratic cancellation.
The formula requires adiabatic control and small accumulated perturbations;
a displayed value between 0 and 1 does not by itself guarantee validity.